

Research Artifact and Reproducibility Package: Objective Clock Bases from Redundant Records

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Artifact report frozen on 2026-06-21

Abstract

This report documents the public research artifact *Research Artifact and Reproducibility Package: Objective Clock Bases from Redundant Records*. The artifact investigates a narrow question at the boundary between quantum Darwinism, redundant records, and relational clock structure. Here a *clock* is not a laboratory timing device; it is a candidate quantum subsystem whose basis labels are interpreted as possible time labels inside a larger quantum state. The central question is: if records make such a clock basis objective, what additional structure is required to recover event order and temporal orientation? The package supports a deliberately limited answer. Fixed-partition redundant records can identify an objective record basis under assumptions already treated in the literature, but that basis is not a physical time line by itself. Persistent physical trajectories are chains in the record-dominance poset. Branching linear extensions are scheduler totalizations, not physical histories. Temporal orientation also requires an anchor beyond redundancy alone. The artifact includes proof notes, exact classical experiments, local quantum-readiness checks, a preregistered four-qubit IBM Quantum illustration on `ibm.kingston`, a locked raw hardware analysis, a secondary readout-mitigation check, a claim-evidence matrix, and a reproducibility archive.

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1 Why This Artifact Exists

The motivating problem comes from quantum foundations rather than from timekeeping engineering. Quantum Darwinism and environment-as-a-witness arguments explain how many environment fragments can carry redundant information about selected system properties [9, 1, 13, 6]. Objective-past work asks how redundant records can support agreement about histories [11]. Fu’s uniqueness theorem clarifies a fixed-partition setting in which a redundantly imprinted observable is unique, while also leaving partition dependence as a boundary [2].

Those results motivate a sharper quantum-clock question. Suppose a larger quantum state contains a candidate clock subsystem C , and suppose independent fragments R_1 and R_2 redundantly record the labels of C . Has that recovered time, or only an objective label basis? The artifact tests the latter view. Agreement on the displayed alternatives is not yet agreement on a physical history. A list of objective quantum snapshots still leaves three distinct tasks:

1. identify the recorded basis;
2. order the recorded snapshots into persistent trajectories;
3. orient the order as earlier-to-later rather than later-to-earlier.

The artifact was built to keep these tasks separate. This matters because a single mathematical object, such as a total ordering of snapshots, can look like a time line while failing the physical persistence condition. The central correction made during the work is that linear extensions of a branching record-poset are not physical time lines. They are merely total schedules compatible with a partial order. A physical scalar time is available only when the relevant snapshots lie on one chain.

For a non-specialist reader, the simplest analogy is a book with copied page notes. The candidate clock label is like a page number. The record fragments are like independent notes copied from the page. Agreement between notes can make the page number objective, but it does not prove that the pages form one story, that the story has no branches, or that the reading direction has been selected by the records themselves. The artifact formalizes this gap in a small quantum setting.

2 Experimental Quantum System

The bounded hardware illustration uses four logical qubits. C is the candidate clock-label subsystem, S is a companion system qubit used in the global coherence witness, and R_1, R_2 are independently accessible record fragments. The GHZ state tests redundant Z -basis records of C in R_1 and R_2 . A matched dephased-control state preserves those local Z records but removes global X coherence. This is the operational reason for measuring both local Z -record correlations and global $XXXX$ -type coherence witnesses.

3 Scientific Boundary

The artifact is not a claim that fixed-partition basis uniqueness is new. It treats pointer-basis and redundant-imprint results as background [12, 2]. It is also consistent with the objective-past program of Riedel, Zurek, and Zwolak, where redundant records of histories support objective past structure [11]. The contribution here is the separation package:

- basis objectivity is not scalar time;
- scalar time requires a chain condition in the record-dominance poset;
- orientation requires an external or physical anchor;
- record capacity and no-go examples make the limits explicit;
- the IBM Quantum run is a bounded illustration, not the proof base.

4 Concept Primer

Concept	Working meaning in the artifact
Clock basis	A basis of a candidate quantum subsystem whose labels are possible time labels; it is not a literal clock face or external timing device.
Objective basis	A basis whose alternatives are redundantly recoverable from independently accessible record fragments under a fixed admissible partition.
Record signature	The tuple of record values associated with a snapshot. In the binary examples, this is a string such as 00, 10, or 11.
Record-dominance poset	The partial order induced by persistence: one snapshot can follow another only when existing records are not lost.
Chain	A totally ordered subset of the poset. Chains are the candidate persistent trajectories.
Linear extension	Any total ordering compatible with the poset. In a branching poset, this is a scheduler artifact, not a physical trajectory.
Scalar time identifiable	The relevant snapshots form one non-duplicated chain, so a one-dimensional timeline can visit all snapshots without record loss.
Orientation anchor	Extra structure that decides which direction of a chain is earlier-to-later.

Table 1: Non-specialist vocabulary used throughout the artifact.

5 Related Work and Positioning

The artifact sits near three established literatures. First, environment as a witness and quantum Darwinism explain how many environment fragments can carry redundant information about selected system properties [9, 1, 13]. Second, spectrum broadcast structure and strong quantum Darwinism formalize stronger notions of objectivity and independence [7, 3, 8, 6]. Third, relational-clock and timeless formulations show why time should not be introduced carelessly as an external primitive [10].

The narrow novelty is not that redundancy can select a basis. Fu’s uniqueness result already addresses uniqueness of the observable leaving redundant imprints within a fixed subset of

Everyday analogy	Quantum object	Interpretation
Page number	C , the candidate clock-label subsystem	The label we are tempted to interpret as time.
Independent copied notes	R_1 and R_2 , record fragments	Separate fragments should recover the same label.
Readable story	Chain in the record-dominance poset	A timeline cannot jump between incompatible branches.
Reading direction	Orientation anchor	Records can show connection without selecting earlier-to-later.
Hidden binding across copies	Global coherence witness	This is a quantum witness distinct from local record readability.

Table 2: Plain-language map from the analogy to the quantum model.

partitions, while also emphasizing ambiguity under different partitions [2]. The artifact instead asks what is still missing after such a basis-level result is granted. The answer is order and orientation.

6 Artifact Architecture

The repository is organized as a gated research pipeline. Each stage writes tracked artifacts with SHA-256 manifests where appropriate. Raw hardware payloads are write-once and secret-free. QPU execution is downstream of the theory and classical checks.

Gate	Role	Representative evidence
G0–G1	Repository and theorem primitives	<code>results/processed/T101_chain_verification.json</code> , <code>results/processed/T103_capacity.json</code> .
G2	Classical C0–C5 experiments	Basis landscape, order catalog, noise phase diagram, assumption failures, exact GHZ analysis.
G3	Local quantum readiness	Circuit manifest, Aer noise sweep, backend-derived local twin.
G4	Frozen preregistration	<code>docs/PREREGISTRATION_FROZEN.md</code> and the frozen circuit packet.
G5	Hardware observation	One completed IBM run on <code>ibm.kingston</code> ; raw H502 primary analysis and H503 secondary mitigation.
G6	Interpretation and packaging	Claim-evidence matrix, final decision, reproducibility archive.

Code family	Meaning	Example
T	Theorem and property checks	T101 verifies the chain criterion.
C	Classical or exact computational experiments	C1 tests basis ambiguity.
Q	Local quantum-readiness checks	Q302 stress-tests the protocol in simulation.
H	IBM hardware execution and analysis	H501 is the QPU job; H502 is raw analysis.
A	Audit, interpretation, and packaging	A603 records the reproducibility archive.
G	Workflow gates	G2 means classical experiments are complete.

7 Core Theoretical Result

The record signatures induce a partial order: a later snapshot may contain more persistent records, but it should not erase records already present. The artifact checks the following operational consequence:

$$\text{scalar time exists} \iff \text{all relevant nonduplicate signatures form one chain.}$$

The theorem-check artifact `results/processed/T101_chain_verification.json` reports 272 record codes checked and 188,158 sequences checked with zero mismatches. The diamond counterexample is decisive: it has two scheduler linear extensions and two maximal persistent chains, but no full persistent scalar order visiting all four snapshots. This is why the report and README avoid calling branching linear extensions physical time lines.

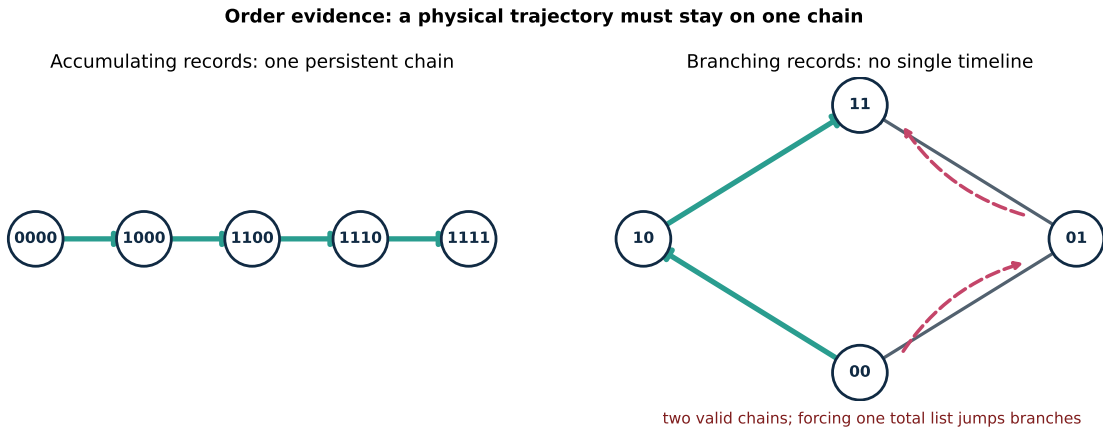


Figure 1: How to read record order. Each circle is a record pattern and each arrow means that a later pattern can keep all records already present. The left example forms one chain, so it can be read as a scalar timeline. The right example branches; forcing all nodes into one list would jump between branches and is not a physical trajectory in this model.

8 Classical Experiments

8.1 Basis Landscape

C1 compares one-record ambiguity with two-record objectivity. The Bell one-record example remains maximally ambiguous in the sampled basis landscape, whereas the GHZ two-record example selects the expected Z -record basis. The C1 summary records $\text{GHZ}_Z = 1.0$ and $\text{GHZ}_X = 0.0$, while the Bell one-record slice has minimum and maximum score 1.0.

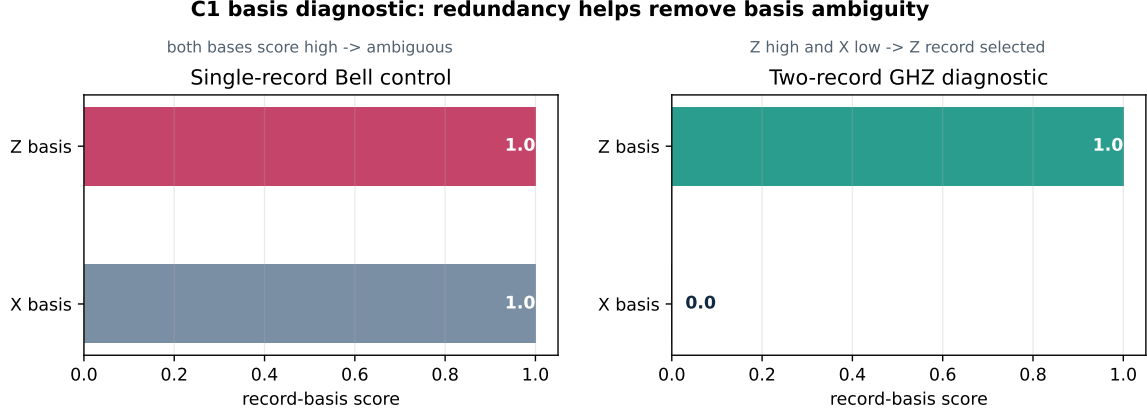


Figure 2: C1 basis diagnostic. A single record can make more than one basis look equally good, which means the basis is ambiguous. With two GHZ record fragments, the Z basis scores high while the X basis scores low, so the record basis is selected under the fixed-partition assumptions.

8.2 Record Capacity

T103 and the C0/C2 family record the capacity constraint for strict-chain binary records. With E binary persistent record coordinates, the maximum strict-chain length is $E+1$. The generated table verifies this from $E = 0$ through $E = 8$, and the boundary check $E = N - 2$ fails for $N = 2$ through $N = 9$, as expected.

8.3 Order Catalog and No-Go Examples

C2 stores six canonical order examples. The thermometer case is classified as `scalar.time`. Diamond and disconnected examples are classified as branching or partial order. Duplicate labels are non-identifiable, erasure is a persistence-violation control, and the orientation-free example records that direction is not identified.

Example	Classification
thermometer	scalar time
diamond	branching or partial order
disconnected	branching or partial order
duplicate	non-identifiable duplicate labels
erasure	persistence violation control
orientation-free	orientation not identified

Table 3: C2 order-catalog summary.

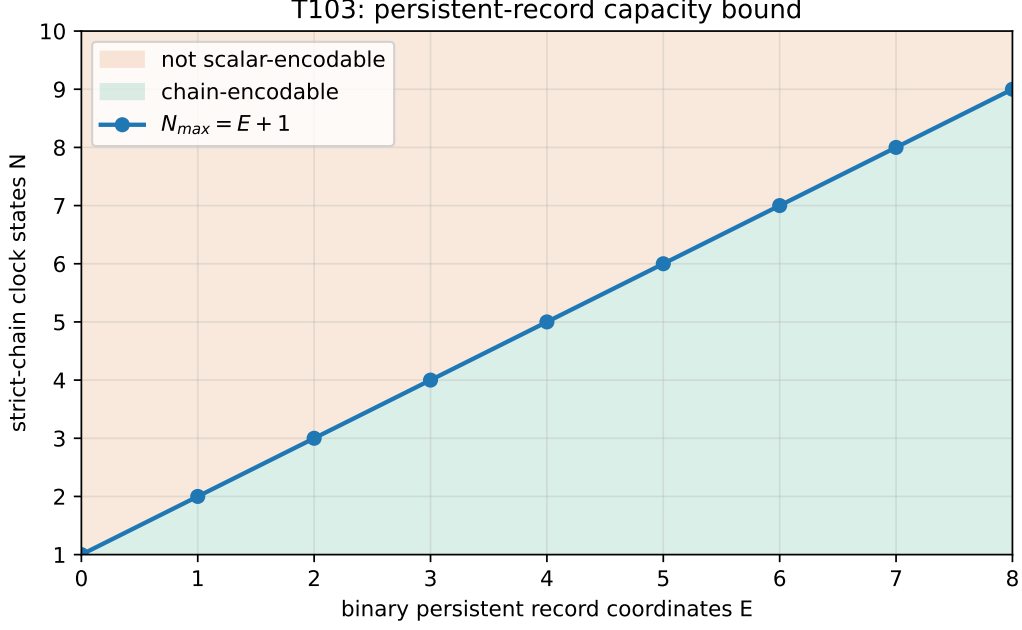


Figure 3: Persistent record capacity. A binary record coordinate is like one persistent switch. With E such switches, at most $E + 1$ strict-chain clock states can be encoded; the shaded region above the line is not scalar encodable.

8.4 Noise and Assumption Stress Tests

C3 evaluates a 112-cell noise grid and records zero bound violations. The recovery boundary moves as expected with bit-flip probability and redundant copies. C4 records assumption failures for label permutations, orientation swaps, unrestricted partitions, single-fragment Bell ambiguity, and diamond branching. These examples define where the positive claims stop.

8.5 Exact GHZ and Dephased-Control Check

C5 checks the exact four-qubit GHZ/dephased-control observables. Both GHZ and dephased controls have Z -record objectivity score 1.0. The coherence witness separates them: GHZ has $\langle XXXX \rangle = 1.0$, while the dephased control has 0.0. These binary exact values are kept as table evidence rather than as a standalone publication figure.

Observable	GHZ	Dephased control
$ZZ_{C,R1}$	1.0	1.0
$ZZ_{C,R2}$	1.0	1.0
$XX_{C,R1}$	0.0	0.0
$XX_{C,R2}$	0.0	0.0
$XXXX$	1.0	0.0

Table 4: C5 exact GHZ/dephased-control sanity check. The control has the same local Z records but no global X -parity coherence.

9 Quantum Protocol and Hardware Observation

The quantum path is deliberately downstream of the theory. Qiskit is used for circuit generation and IBM Runtime submission, with a SamplerV2-style bitstring workflow consistent with the

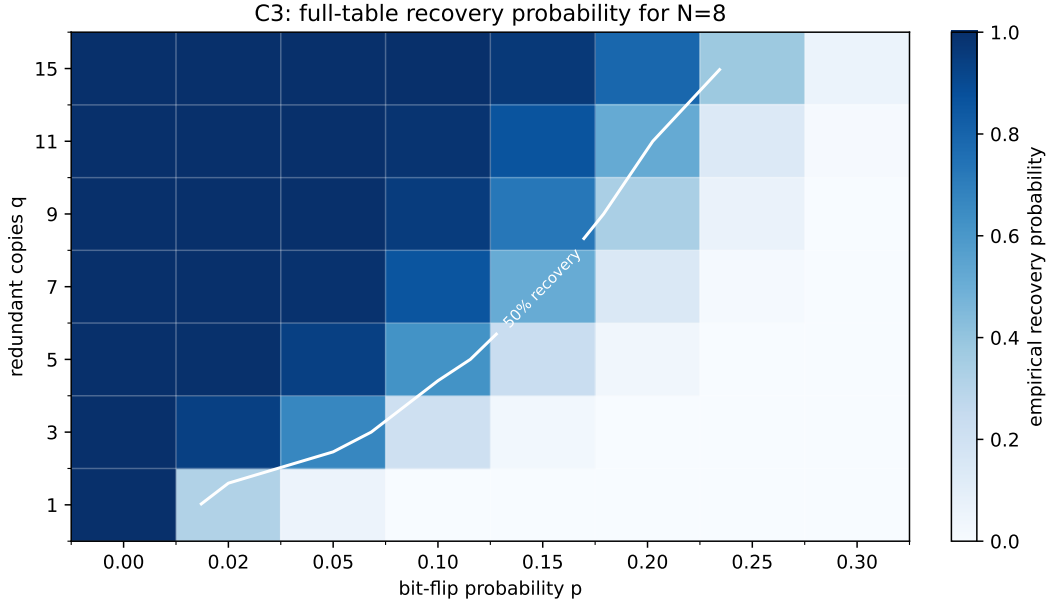


Figure 4: C3 noise phase diagram. Darker blue cells mean the simulated record table is recovered more often, pale cells fail more often, and the white contour marks the approximate 50 percent recovery boundary. More redundant copies improve recovery, but no noise result removes the chain and orientation requirements.

Runtime primitive documentation [5, 4]. The preregistered packet contains 24 circuit instances and 24,576 total shots. The first submitted `ibm_marrakesh` job was cancelled while queued and is recorded only as pre-result provenance. After explicit approval, a pre-result backend amendment regenerated the dependent artifacts for `ibm.kingston`.

Field	Value
Backend	<code>ibm_kingston</code>
Job id	<code>d8rfasuab0ds73drkaig</code>
Status	DONE
Created	2026-06-20T20:17:55.809388Z
Running	2026-06-20T20:17:57.209474Z
Finished	2026-06-20T20:18:10.354081Z
Circuit instances	24
Observed shots	24,576
Quantum usage	9 seconds
Secret values recorded	false

Table 5: Completed H501 hardware observation.

The secondary H503 readout-assignment correction agrees qualitatively with the raw result. Its tensor assignment condition number is 1.1172921620957568, with corrected $\Delta_{\text{obj}} = 0.9850328718815358$ and corrected $\Delta_{\text{coh}} = 0.9687926905262139$. This check cannot rescue a failed raw result; it only records agreement with the preregistered primary decision.

10 Claim Discipline

A601 records ten claims, zero unsupported claims, and two Q-grade hardware illustration claims. The final artifact route is `foundations.artifact_with.bounded.quantum.illustration`: the

Raw H502 metric	Value
Raw primary inclusion pass	true
Δ_{obj} lower confidence bound	0.8876953125
Minimum Z-correlation lower confidence bound	0.9169542107266444
Δ_{coh} lower confidence bound	0.877685546875
$\langle XXXX \rangle_{\text{plus}}$	0.8955078125
$\langle XXXX \rangle_{\text{minus}}$	-0.87744140625
$\langle XXXX \rangle_{\text{mix}}$	0.009033203125
Bootstrap seed	20260621
Bootstrap replicates	10,000

Table 6: Locked raw hardware analysis. These are the primary values used for artifact inclusion.

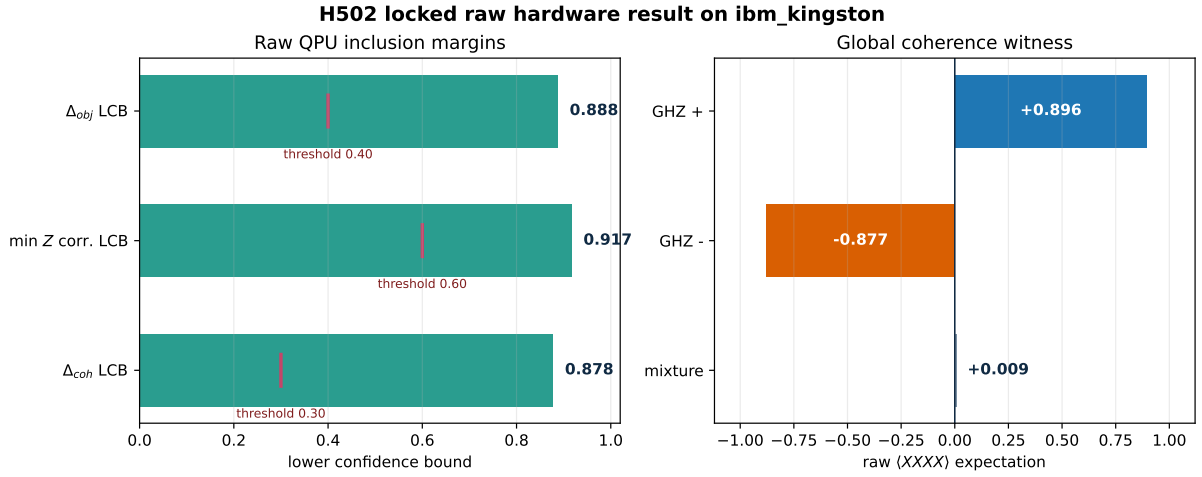


Figure 5: H502 locked raw hardware result. The left panel compares raw lower confidence bounds with preregistered pass/fail thresholds. The right panel shows the measured global $XXXX$ coherence witness: the GHZ-plus and GHZ-minus circuits separate from the near-zero mixture.

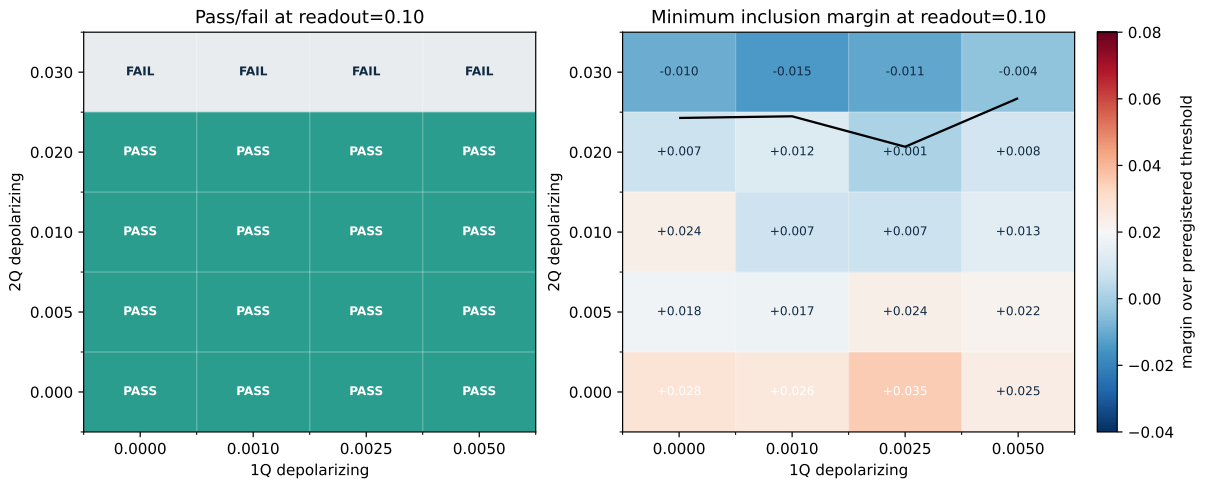


Figure 6: Q302 local readiness sweep before QPU execution. The left panel shows pass/fail status at the highest readout-flip setting in the sweep. The right panel shows the numerical margin over the preregistered thresholds at that same readout setting. The hardware run is interpreted only after theory, local simulation, preregistration, and raw analysis gates have passed.

theory package is supported, and the hardware observation is eligible only as a bounded quantum illustration.

Positive claim	Boundary
Basis-order-orientation separation	Supported by proofs, exact checks, and counterexamples.
Record capacity and no-go package	Supported independently of QPU data.
IBM hardware result	Supports a bounded four-qubit GHZ illustration only.

Table 7: Allowed positive interpretations.

Rejected interpretation	Reason
Basis uniqueness as the main novelty	This is imported context under fixed-partition assumptions.
Hardware proof of the theorem package	The theory and no-go results are established upstream of H501.
Orientation from redundancy alone	Orientation-swap examples remain valid without an anchor.
Mitigation replacing raw analysis	H503 is secondary and cannot override H502.

Table 8: Explicit negative interpretations frozen by A602.

11 Reproducibility

The frozen reproducibility archive is `objective-clocks-reproducibility.tar.gz`. Its SHA-256 is recorded outside the archive payload in `results/processed/A603_reproducibility_manifest.json` and `docs/FINAL_DECISION.md`. This avoids embedding the archive hash inside files that are themselves archived. The A603 manifest records the archived file set, source commit, a successful clean reproduction run, and no recorded secret values. The clean reproduction command was:

```
python scripts/reproduce_all.py
```

The recorded terminal tail ends with 55 passing tests. The public repository also includes `CITATION.cff`; until a manuscript is released, the artifact should be cited by title, author, repository URL, frozen tag, and the manifest archive hash.

12 Limitations

The artifact is intentionally small in hardware scale. The IBM run is a four-qubit illustration of the GHZ/dephased-control workflow, not a many-body demonstration of objective histories. The theoretical claims depend on the specified record model and admissible partitions. The report does not solve partition-independent Hilbert-space factorization or derive an orientation anchor from redundancy alone. Those remain outside the claim boundary.

A Three-Round Review Record

The final report and README were reviewed through four roles across three rounds. The review was performed as an internal artifact audit rather than as peer review.

Round	Reviewer role	Challenge	Resolution
1	Conceptual reviewer	The README and report could overstate basis uniqueness as novelty.	Moved fixed-partition basis uniqueness to related-work context and made the basis-order-orientation separation the central claim.
1	Mathematical reviewer	Linear extensions might be mistaken for physical histories.	Promoted the chain criterion and diamond counterexample to the core result.
1	Hardware reviewer	The IBM result could be read as proof of the theory.	Labeled H501 as a bounded illustration and kept raw H502 as primary.
1	Reproducibility reviewer	The artifact needed a direct archive/hash story.	Added the archive SHA-256, source commit, file count, and reproduction command.
2	Conceptual reviewer	Non-specialists need vocabulary before the tables.	Added a concept-primer table before technical evidence.
2	Mathematical reviewer	The orientation no-go boundary needed explicit wording.	Added orientation-anchor language to the abstract, primer, and limitations.
2	Hardware reviewer	Backend amendment provenance could look like post-result selection.	Documented the cancelled queued Marrakesh job as pre-result provenance and identified Kingston as the only completed observation.
2	Reproducibility reviewer	Operational details should not dominate the scientific changelog.	Moved changelog wording toward scientific chronology and left low-level details to manifests and docs.
3	Conceptual reviewer	The motivation should explain why clocks require more than objectivity.	Rewrote the opening around agreement on alternatives versus agreement on a physical history.
3	Mathematical reviewer	Capacity and no-go claims needed evidence-file pointers.	Added T101, T103, C2, C3, C4, and C5 references in the methods narrative.
3	Hardware reviewer	Mitigation status needed stronger guardrails.	Stated that H503 cannot rescue or replace H502.

Round	Reviewer role	Challenge	Resolution
3	Reproducibility reviewer	Figures needed visual-context justification.	Included the tracked publication figures and described what each one supports.

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